NetSilicon® NS9360

Overview | Specs | Documentation

32-bit ARM926EJ-S RISC Processor

- 103, 155, 177 MHz
- 5-stage pipeline
- Harvard architecture
- 8 kB I-cache and 4 kB D-cache
- 32-bit ARM and 16-bit Thumb instruction sets, can be mixed for performance/code density tradeoffs
- MMU to support virtual memory based OS's such as Linux,WinCE/Pocket PC, VxWorks, etc.
- DSP instruction extensions: improveddivide, single cycle multiplyaccumulate
- ARM Jazelle, 1060CM (CaffeineMarks) Java Accelerator
- Embedded ICE-RT debug unit
- JTAG boundary scan support
- Clock-gated processor for decreased power dissipation

External System-Bus Interface

- 32-bit data bus, 28-bit external address bus
- Glueless interface to SDRAM,SRAM, EEPROM, buffered DIMM, Flash
- Up to 256 MB SDRAM, up to 2 GB DIMM
- 4 static and 4 dynamic chip selects
- 0-63 wait states per chip select

- Self-refresh during system sleep
- Automatic dynamic bus sizing to 8-bits, 16-bits, 32-bits
- Burst-mode support with automatic data width adjustment
- 2 external DMA channels for external peripheral support

System Boot

- High-speed boot from 8-bit, 16-bit,or 32-bit ROM or Flash
- Hardware-supported low cost boot from serial EEPROM through SPI port (patent pending)

Optimized 10/100 Ethernet MAC

- MII or RMII PHY interfaces
- Full- or half-duplex
- Station, broadcast, multicast address filtering
- 2 kB Rx FIFO
- 256B Tx FIFO with on-chip buffer descriptor ring (eliminates under runs and decreases bus traffic)
- Separate Tx and Rx DMA channels
- Intelligent receive-side buffer sizeselection
- Support for full statistics gathering
- Support for external CAM filtering

Flexible LCD Controller

- Supports commercially available displays up to SVGA
- Active Matrix color TFT displays
- Up to 18 bpp; 256k colors
- Single and dual-panel color Passive Matrix displays
  - Up to 16 bpp 4:4:4 RGB; 3375 colors
- Single and dual-panel monochrome STN displays
- 1, 2, 4 bpp palletized grayscale
- Formats image data and generates timing control signals
- Internal programmable palette-LUT and grayscaler support different color techniques
- Programmable panel-clock frequency

USB Ports

- USB v.2.0 Full Speed (12 Mbps) and Low Speed (1.5 Mbps)
- OHCI host and 13 end point device
- Single PHY can be used with either host or device
- Interface to external PHY for simultaneous host and device operation
- USB host is a bus master
- Each USB device endpoint is supported by a dedicated DMA channel, 13 total
- 20B Rx FIFO and 20B Tx FIFO

Serial Ports

- 4 serial modules, each independently configurable to UART mode, SPI master mode, or SPI slave mode

- Bit rates from 75 bps to 1.8 Mbps:asynchronous x8 mode
- Max bit rates for synchronousmode are:
  - •1/16 CPU speed for SPI master
  - •1/32 CPU speed for SPI slave
- UART provides:
  - •High-performance hardware and software flow control
  - •Odd, even, or no parity- 5, 6, 7 or 8 bits
  - •1 or 2 stop bits
  - •Receive-side character and buffer gap timers
- Internal or external clock support for synchronous mode
- 4 receive-side data match detectors
- 2 dedicated DMA channels per module, 8 total
- 32B Tx FIFO and 32B Rx FIFO per module

I2C Port

- I^2C v.1.0, configurable to master or slave mode
- Bit rates: fast (400 kHz) or normal (100 kHz) with clock stretching
- 7-bit and 10-bit address modes

1284 Parallel Peripheral-to-Host Port

- All standard modes:
  - •ECP, Byte, Nibble, Compatibility
- RLE (Run Length Encoding) decoding of compressed data in ECP mode
- Operating clock from 100 kHz to 2 MHz
- 4 dedicated DMA channels
  - •2 for data and 2 for control
- Microsoft Plug-and-Play, no Windows driver needed

System Bus DMA

- Every system bus peripheral is a bus master with dedicated DMA engine
- Deterministic bus bandwidth allocation (patent pending)

Peripheral Bus DMA

- 13-channel engine supports USB Device
  - •2 DMA channels support control endpoint
  - •11 DMA channels support 11 endpoints
- 12-channel engine supports
  - •4 serial modules (8 DMA channels)
  - •1284 parallel port (4 DMA channels)
- All DMA channels support fly-by mode

External Peripheral DMA

- 2-channel DMA engine
- Supports memory-to-memory transfers

Power Management

- Power save during normal operation
- Disables unused modules
- Power save during sleep mode
  - Sets SDRAM to self-refresh mode
  - Individually disables every module except selected wakeup modules
  - Wakeup on valid packets or characters
- Patent pending technology

Vectored Interrupt Controller

- Holds pointers to all interrupt service routines for rapid service
- Services all peripherals
- Hardware interrupt prioritization

General Purpose Timers/Counters/PWM

- 8 independent 16- or 32-bit programmable timers, counters, or 4PWM functions
- Each has an I/O pin
- Mode selectable into:
  - Internal timer mode
  - External gated timer mode
  - External event counter
  - PWM
- Timers/counters can be concatenated
- Minute-range events measurable
- Source clock selectable
- Internal clock or external pulse event
- Individually enabled/disabled

System Timers

- Watchdog timer
- System bus monitor timer
- Peripheral bus monitor timer

General Purpose I/O

- 73 programmable GPIO pins (muxed with other functions)
  - includes 7 high-current (8mA) GPIO pins
- Software-readable power-up status registers for customer-defined bootstrapping

External Interrupts

- 4 external programmable interrupts
  - Rising or falling – edge sensitive
  - Low or high-level sensitive

Real Time Clock

- Time of day clock
- Alarm
- 100 year calendar

- Programmable periodic interrupt
- 10 ms resolution
- Dedicated time domain in the system PLL
  - •RTC-only mode available
- Initial time from network through SNTP routine
  - •No battery backup
- Additional benefits
  - •Frees CPU from math calculations
  - •Decreased response time for queries

Clock Generator

- Low cost external crystal
- Internal phase locked loop (PLL)
- Software programmable PLL parameters
- Optional external oscillator
- Separate oscillator for USB

Operating Voltage

- Core: 1.5V ± 0.1V
- I/O ring: 3.3V ± 10%

Operating Frequency

- 103 MHz: 0 °C to 70 °C
- 155 MHz: -40 °C to +85 °C
- 177 MHz: 0 °C to 70 °C

Power Consumption

- 177 MHz: 1.35W max (estimated)

Package

- 272-pin BGA including 16 thermal balls
- 1.27 mm ball pitch
- 27 mm x 27 mm
- Lead free, RoHS compliant

Details on ARM9 Core

- 32/16-bit Harvard RISC architecture with 5 stage pipeline (ARMv5TEJ)
- 32-bit ARM instruction set for maximum performance and flexibility
- 16-bit Thumb instruction set for increased up to 35% code density
- DSP instruction extensions and single cycle MAC
- ARM Jazelle technology, 6 CM/MHz = 1200 Caffeine Marks @ 200 MHz
- MMU which supports Windows CE and Linux
- EmbeddedICE-RT logic for real-time debug